THE PROJECTION FORMULA FOR CAPPING WITH A CHERN CLASS AND FLAT PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The characterisation of the Chern classes by one relation on the projective bundle. Same chapter chow.tex, label lemma-determine-intersections, invoked at proof lines 7454 and 7480. Provides the characterising relation for the classes obtained by capping with the Chern classes, and the uniqueness that identifies the classes obtained after pulling back.
- Compatibility of the cap with a first Chern class and with flat pullback, and preservation of rational equivalence. Same chapter chow.tex, labels lemma-flat-pullback-cap-c1 (proof line 7473), lemma-compose-flat-pullback (proof line 7474) and lemma-flat-pullback-rational-equivalence (proof line 7472). They let the relation be pulled back along the morphism between the projective bundles and rewritten in terms of the pulled back classes.

Source reference. Original lemma 02RD: sources/stacks-chow/pinned-original-source.tex, source lines 2287–2287. Navigation pointer for \ref{lemma-compose-flat-pullback} used inside the counted span (line 7474); the argument that consumes it is retained in the span.

Source reference. Original lemma 02U6: sources/stacks-chow/pinned-original-source.tex, source lines 7280–7280. Navigation pointer for \ref{lemma-determine-intersections} used inside the counted span (line 7454); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02SS: sources/stacks-chow/pinned-original-source.tex, source lines 4533–4533. Navigation pointer for \ref{lemma-flat-pullback-cap-c1} used inside the counted span (line 7473); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S1: sources/stacks-chow/pinned-original-source.tex, source lines 3173–3173. Navigation pointer for \ref{lemma-flat-pullback-rational-equivalence} used inside the counted span (line 7472); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 7441); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $\mathcal{E}$ be a finite locally free sheaf of rank r on Y. Let $f : X \to Y$ be a flat morphism of relative dimension r. Let α be a k-cycle on Y. Then*

$$f^*(c_j(\mathcal{E}) \cap \alpha) = c_j(f^*\mathcal{E}) \cap f^*\alpha$$

Proof. Write $\alpha_j = c_j(\mathcal{E}) \cap \alpha$, so $\alpha_0 = \alpha$. By Lemma 02U6 we have

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_P(1))^i \cap \pi^*(\alpha_{r-i}) = 0$$

in the chow group of the projective bundle $(\pi : P \to Y, \mathcal{O}_P(1))$ associated to $\mathcal{E}$. Consider the fibre product diagram

$$\begin{array}{ccc} P_X = \mathbf{P}(f^*\mathcal{E}) & \xrightarrow{f'} & P \\ {\scriptstyle \pi_X}\downarrow & & \downarrow{\scriptstyle \pi} \\ X & \xrightarrow{f} & Y \end{array}$$

Note that $\mathcal{O}_{P_X}(1)$ is the pullback of $\mathcal{O}_P(1)$. Apply flat pullback $(f')^*$ (Lemma 02S1) to the displayed equation above. By Lemmas 02SS and 02RD we see that

$$\sum\nolimits_{i=0}^{r} (-1)^i c_1(\mathcal{O}_{P_X}(1))^i \cap \pi_X^*(f^*\alpha_{r-i}) = 0$$

holds in the chow group of P_X. By the characterization of Lemma 02U6 we conclude. □